

Utkarsh Bharadwaj

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About Me

Mathematics and statistics student at the Indian Statistical Institute, Delhi, passionate about applying probabilistic and optimization frameworks to real-world problems. Eager to contribute to quantitative research and data-driven roles.

Education

Indian Statistical Institute

Bachelor of Statistical Data Science (BSDS)

Delhi, India

Aug 2025 – Present

- Score: **70.4%** (1st Semester); **84%** (2nd Semester)
- **Coursework:** R Programming, Linear Algebra, Probability Theory, Real Analysis, Statistical Inference, Hypothesis Testing, Linear Regression, Sampling Theory, Optimization Theory and Numerical Methods

Technical Skills

Languages: Python, R, $\text{\LaTeX}$

Libraries & Frameworks: NumPy, ggplot2, lattice

Statistical Methods: Poisson Processes, QP/Convex Optimization, Benford's Law, Regression Analysis, Hypothesis Testing, Interval Estimation

Tools & Environment: Linux (Debian), VS Code, RStudio, GitHub

Projects

Neyman-Pearson Lab: Hypothesis Testing Tool | *Hypothesis Testing, MP Test* July 2026

- Built a single-page interactive web application that derives Most Powerful (Neyman-Pearson), UMP (Karlin-Rubin), and UMPU level- α tests live for nine one-parameter distribution families.
- Designed a unified derivation engine producing closed-form critical regions and step-by-step, KaTeX-rendered proofs of the rejection region for user-specified hypotheses and significance levels.

CipherInk: Forensic Stylometry Platform | *Bayesian Inference, Statistical NLP* June 2026

- Developed a full-stack, multi-author forensic stylometry web application for authorship attribution.
- Engineered a backend using Poisson log-likelihood and Bayesian softmax to compare statistical fingerprints of 70 filler words against a calibrated global English baseline.

Beyond the Deficit: India's Trade Dynamics | *R, Time-Series Analysis, Macroeconomics* May 2026

- Conducted a 35-year empirical time-series analysis (1990–2025) of India's macroeconomic trade trends.

- Evaluated structural impacts of currency depreciation and energy dependency using inflation-adjusted (real) values.

Convex Optimization for LPG Allocation | *QP, Optimization Theory* *Apr 2026 – May 2026*

- Formulated a strictly convex Quadratic Programming (QP) model to optimally allocate constrained national fuel supply across domestic and commercial sectors.
- Designed an objective function minimizing the weighted sum of squared shortfalls to balance commercial economic losses against environmental health risks, using historical consumption data and demographic vulnerability weights.

Certifications & Workshops

Indian Institute of Technology Ropar

NS2ML-DE 2026 Workshop

Ropar, India

Jul 6 – 10, 2026

- Covered numerical PDE/ODE solvers (finite difference/element methods) and data-driven ML techniques (Physics-Informed Neural Networks), with hands-on MATLAB/Python labs.

SQC & OR Unit Seminar

Extremal Bayesian Deep Learning for Weather Extremes

ISI Delhi, India

Jul 16, 2026

- Attended a talk by Dr. Rishikesh Yadav (Assistant Professor, IIT Mandi) on a Bayesian deep learning framework for prediction and uncertainty quantification of weather extremes.
- Got a solid introduction to Extreme Value Theory - the GEV and GPD distributions, and the block-maxima vs. peak-over-threshold approaches to modeling extremes and their applications in forecasting weather and climate extremes.

Achievements

ISI Merit List: Secured All India Rank **31** for the BSDS program at the Indian Statistical Institute.

JEE Main 2025: Secured **99.37 percentile** in Mathematics, placing in the top 0.6% of applicants nationwide.

Vihaan 9.0 Hackathon Finalist: Advanced to the offline finals at Delhi Technological University with Team Diazonium.

NIT Delhi UpVision Hackathon: Placed Top 10 of 150+ teams in the “AI for Humanity” track with Team Diazonium.

Languages

Maithili (Mother Tongue), Hindi (Native Speaker), English (Advanced – IELTS Certified, C1 Level).